Original Research

Vaginal Estrogen Application Techniques for Prevention of Urinary Tract Infection

A Randomized Trial

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OBJECTIVE: To determine whether periurethral estrogen application is noninferior to intravaginal estrogen application for the prevention of urinary tract infections (UTIs) in postmenopausal individuals with laboratory-proven recurrent UTI.

METHODS: The TAPER (Techniques of Applying Vaginal Estrogen for Prevention of Recurrent Urinary Tract Infections) trial was a single-center, nonblinded, randomized noninferiority trial that compared outcomes after a twice-weekly intravaginal estradiol applicator instillation of 1.0 g cream (intravaginal) compared with periurethral digital application of 0.5 g cream (periurethral) in postmenopausal individuals with recurrent UTIs. The primary outcome was the proportion of UTI-free participants at 6 months postintervention. Secondary outcomes included number of UTIs, the amount of time elapsed to the first UTI, vaginal pH, PGI-I (Patient Global Impression of Improvement) scores, patient experience with the study drug, study drug adverse effects, and urinary and sexual function. To achieve 80% power with a type 1 error of 5% and to account for loss to follow-up, 114 participants needed to be randomized based on a noninferiority margin of 25%.

RESULTS: Of 270 eligible individuals, 114 consented and were randomized (57 intravaginal, 57 periurethral). Participants were predominantly White (92.1%) with a mean age of 71.1 years (range 52–93). Participants reported a median of three UTIs in the 12 months before enrollment. Individuals randomized to periurethral estrogen application had a similar proportion of UTI-free participants at 6 months as those randomized to intravaginal estrogen application (50.9% vs 52.6%; risk difference, −1.75 percentage points; 95% CI, −20.0 to 17.0 percentage points). Both groups experienced a decrease in vaginal pH at 6 months; the decreases did not differ by application method (−1.2 intravaginal vs −0.8 periurethral, *P*=.30). Participants assigned to intravaginal estrogen were more likely to experience vaginal itching after 3 months of use than those assigned to periurethral application (24.0% vs 2.4%, *P*<.01). Other outcomes did not differ by route of estrogen application.

CONCLUSION: For postmenopausal individuals who are experiencing recurrent UTI, periurethral application of estradiol cream is noninferior to intravaginal application for UTI prevention. Both application methods

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resulted in approximately half of the participants being UTI-free at 6 months.

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Menopause confers significant risk for recurrent urinary tract infections (UTIs) due to changes in the urogenital epithelium and microbiome from estrogen depletion.[1] Nearly 50% of individuals diagnosed with a UTI develop *recurrent UTIs*, defined as at least two culture-proven UTIs in 6 months or at least three in 1 year.[2] Urinary tract infections increase older adults' risk for serious health consequences, which include sepsis, hospitalization, antibiotic resistance, and death.[3]

Nonantibiotic therapies exist to prevent UTIs, but vaginal estrogen is the only option with a grade A recommendation.[4,5] Vaginal estrogen therapy restores beneficial *Lactobacillus* species within the urogenital microbiome, which promotes an acidic vaginal environment protective against uropathogens.[6,7] It is commonly prescribed as a cream that is instilled intravaginally using a plastic applicator.[8] However, high discontinuation rates (up to 90%) are reported, with users reporting that intravaginal estrogen cream application is messy, not user-friendly, or that it is painful and expensive.[9,10] Some clinicians alternatively recommend digital application of a smaller, pea-sized amount to the periurethral area, despite a lack of published data supporting effectiveness of this method.[11]

We conducted a single-center, nonblinded, randomized noninferiority trial that compared periurethral application of 0.5 g estradiol cream (experimental) with intravaginal application of 1.0 g (standard) estradiol cream for the prevention of UTIs over 6 months in postmenopausal individuals with recurrent UTI.

METHODS

The TAPER (Techniques of Applying Vaginal Estrogen for Prevention of Recurrent Urinary Tract Infections) trial, a single-center, nonblinded, randomized noninferiority trial, was conducted at two hospitals under a tertiary care academic medical center and was approved by the University of Pittsburgh Institutional Review Board. The trial was registered at ClinicalTrials.gov (NCT05472779, July 25, 2022). Details of the study design and methods with detailed inclusion and exclusion criteria have been previously published.[12]

Postmenopausal patients with a history of recurrent UTIs were recruited from urogynecology, urology, women's health, and infectious disease clinics, as well as through other marketing methods (eg, Pitt+Me communication to individuals interested in participation in medical research). *Postmenopausal status* was defined as having no menses for more than 12 months, bilateral oophorectomy, age older than 56 years (95th percentile for age at menopause), or more than 30 international units/L of follicle-stimulating hormone in patients who were medically or surgically amenorrheic. *Urinary tract infections* were defined as the presence of symptoms (eg, dysuria, urinary frequency) and a positive urine culture or urinalysis with positive nitrites, moderate or many bacteria, 2+ large leukocyte esterase, or a combination of these. We required laboratory evidence within the previous year of recurrent UTI (two or more UTIs in the previous 6 months or three or more UTIs in the previous 12 months). Individuals currently taking alternative nonantibiotic medications for UTI prevention were able to enroll in the TAPER trial (eg, methenamine hippurate, cranberry supplementation, D-mannose, and probiotics). We excluded those receiving chronic antibiotic prophylaxis, antibiotic bladder instillations for bladder pain or infection, as well as those exposed to exogenous estrogen in the previous 3 months. Individuals with significant vaginal stenosis or shortening, contraindications to exogenous estrogen, or those using intermittent or indwelling urinary catheterization were also excluded. Enrollment of participants who consented occurred from January 3, 2023, to October 8, 2024.

Randomization was 1:1 to either intravaginal or periurethral application of estradiol cream by using a fixed block size of 4, stratified by clinical site. Randomization allocation was determined using a secure web browser with the uploaded randomization sequence that was created by a statistician who was not involved with study recruitment. Study team personnel did not have access to the random allocation sequence. Participants assigned to the intravaginal group were instructed to use the applicator to instill 1.0 g of 0.01% estradiol cream into the vagina. Participants in the periurethral group were instructed to apply 0.5 g of 0.01% estradiol cream to the periurethral tissues using a finger. The frequency of cream use was the same for both groups: nightly for 14 days, followed by application twice weekly at bedtime. All participants were provided instructions on how to use a calibrated plastic applicator to measure vaginal cream (applicators with a 0.5 g marking were given to those randomized to periurethral administration). Participants were asked to demonstrate their ability to appropriately dispense and measure the study drug by way of the "teach-back" technique.[13] Participants and study team members were not masked to either study arm. After the baseline study visit, participants had 3-month and 6-month visits to capture clinical and patient-reported outcomes and assess study drug adherence. Remote participation in the trial was provided as an option midway through the study to expand enrollment to rural areas. Remotely enrolled participants received the study drug by mail and were taught application techniques over video, with survey completion performed using a secure web application.

The primary trial outcome was the proportion of UTI-free participants 6 months after the initiation of estradiol cream. Any UTI diagnosed within the first 4 weeks of study drug use was not counted towards this outcome, given the known delay of approximately 3–4 weeks in the effects of vaginal estrogen therapy after initiation.[14,15] The primary outcome was assessed by chart review and confirmed by patient report. To reduce recall bias, participants were queried at each research encounter about the occurrence and frequency of UTIs they had experienced in the previous 3 months. The amount of time to first the UTI and number of UTIs were also collected for comparison. Study drug adherence was assessed by estradiol tube weights at baseline, 3 months, and 6 months, and by survey questions about adherence.

Secondary measures included vaginal pH and patient-reported outcomes. Vaginal pH was assessed 2 cm proximal to the vaginal hymen at the posterior vagina using pH paper. Patient-reported outcomes included the UDI-6 (Urogenital Distress Inventory, Short Form) to assess stress and urge urinary incontinence bother and symptom severity,[16] the FSFI-6 (Female Sexual Function Index-6) to assess sexual dysfunction,[17] and a 10-question, nonvalidated survey that assessed patient experience with the study drug using binary (yes or no) or 5-point Likert scale questions. Participants reported adverse effects attributed to use of the study cream at the 3-month and 6-month visits. Additional adverse events were listed in a systematic fashion, and participants were asked whether they believed symptoms were related to study drug use.

Sample size calculations were performed based on a noninferiority test of two independent proportions, with a type 1 error of 5% and 80% power. Using Eriksen et al.'s study, a predicted 45% UTI-free rate at 6 months was applied.[18] Assuming a noninferiority margin of 25% and a 45% UTI-free rate in each arm, 49 participants per arm provided 80% power to demonstrate noninferiority. To account for 14% loss to follow-up, we aimed to enroll 114 participants, 57 per arm.

The primary analysis was an intention-to-treat analysis, with a secondary planned per-protocol analysis that excluded participants who failed to follow their assigned arm for more than 50% of study involvement or those who, over the course of the 6-month study, no longer met eligibility criteria. Participants who were lost to follow-up were included in the per-protocol analysis if there was sufficient documentation in the medical record exhibiting their compliance with the study drug during 6-month follow-up. Data from participants who withdrew from the study were included in final analyses for the primary outcome if they allowed the study team to access the medical record for chart review after withdrawal. Baseline demographic and clinical characteristics were compared with *t* test or Wilcoxon rank-sum tests for continuous variables (based on normality) and χ^2 or Fisher exact tests for categorical variables. Self-reported race and ethnicity data were collected to allow for assessment of study generalizability. Logistic regression was performed for the primary outcome. A multivariable sensitivity analysis was performed for the primary outcome of proportion of UTI-free participants, controlling for any covariate imbalance after randomization. Kaplan-Meier curves with a log-rank test were also performed to compare UTI-free survival between the groups. Group differences in patient-reported outcomes and vaginal pH were explored using two-sample *t* tests for secondary outcomes. Imputation was not conducted for missing data for study outcomes and covariates. Analyses were performed using Stata 16.1.

RESULTS

Of the 270 individuals who were identified as eligible for the study, 104 declined participation and 51 could not be contacted (Fig. 1). Thus, 114 participants were enrolled and randomized, with 57 in the intravaginal arm and 57 in the periurethral arm. Twenty-three (20.2%) participants were remotely enrolled into the study (n=10 intravaginal; n=13 periurethral). The mean age of the cohort was 71.1 (standard deviation [SD]: 8.9) years with an age range of 52–93 years, and the mean body mass index (BMI, calculated as weight in kilograms divided by height in meters squared) was 31.2 (SD 8.3) (Table 1). The majority of participants identified as White (92.1%), followed by Black (6.1%). In both groups, the median number of UTIs in the previous 12 months was three (IQR 2–4 intravaginal vs 3–4 periurethral).

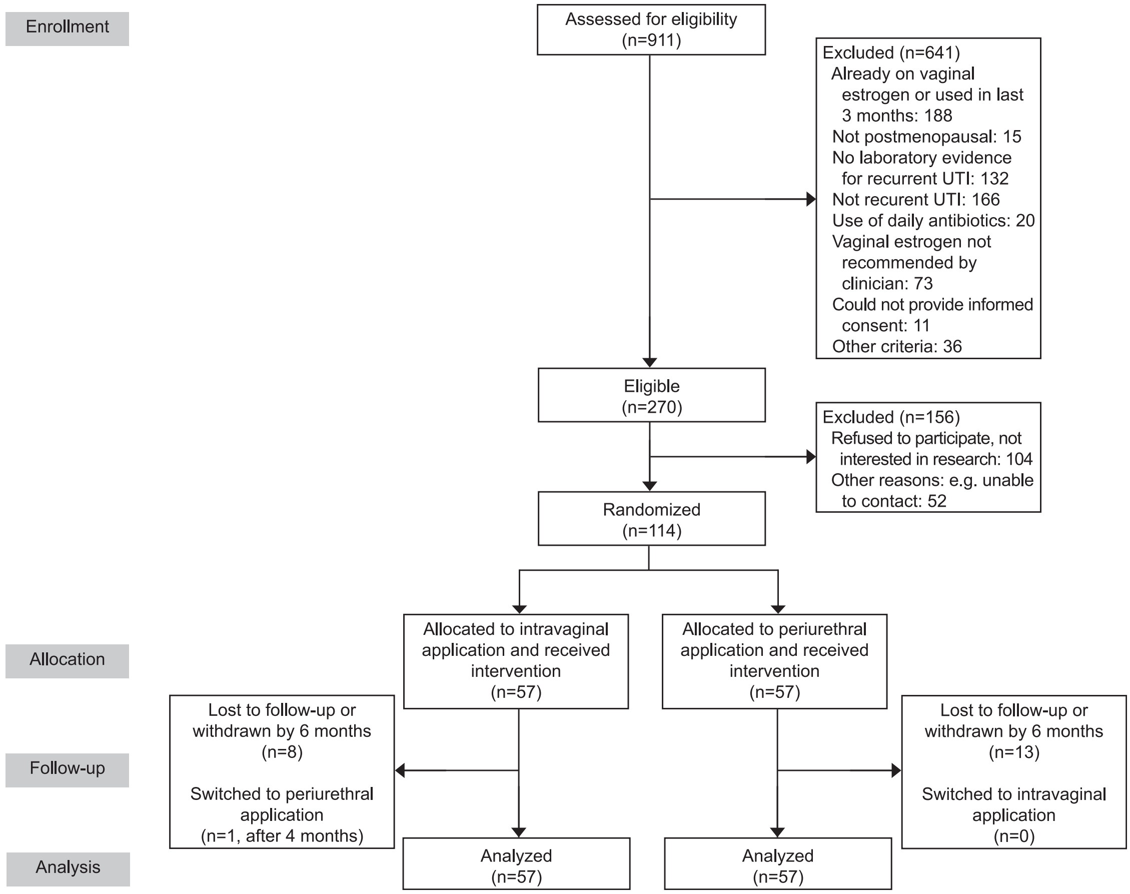

Fig. 1. CONSORT (Consolidated Standards of Reporting Trials) flow diagram. UTI, urinary tract infection.
Zuo. RCT of Topical Estrogen to Prevent UTIs. Obstet Gynecol 2026.

Demographic and clinical characteristics were similar between the arms (Table 1). However, participants in the periurethral application arm were less likely to be sexually active compared with those in the intravaginal application arm (19.6% vs 42.9%, $P<.01$).

The proportion of participants who were UTI-free at 6 months did not differ by estrogen-application method (52.6% intravaginal vs 50.9% periurethral, risk difference −1.75 percentage points; 95% CI, −20.0 to 17.0 percentage points) (Table 2). The median number of UTIs (0 intravaginal [IQR 0–1] vs 0 periurethral [IQR 0–2], $P=.53$) and the time to the first UTI (89 days intravaginal [IQR 55–147] vs 83 days periurethral [IQR 45–121], $P=.08$]) were similar in both groups at 6 months. An adjusted Cox proportional hazards model, controlling for baseline sexual activity, revealed no difference in UTI-free survival between the groups, with administrative censoring at 6 months (hazard ratio 0.93; 95% CI, 0.53–1.63; Fig. 2). A per-protocol analysis was performed to exclude participants with self-reported or documented adherence of less than 50% with the study medication (see Appendix 1, available online at https://links.lww.com/AOG/E759, for per-protocol participant characteristics). Estrogen compliance based on study drug tube weights were considered in the minority of participants (39.5% of in-person participants) who brought the medication at follow-up research visits. The per-protocol analysis also demonstrated no difference in the proportion of UTI-free individuals between periurethral and intravaginal application (risk difference: −3.1 percentage points; 95% CI, −23.0 to 17.0 percentage points). Because the CI includes the prespecified noninferiority margin of 25%, we can conclude noninferiority of periurethral estradiol administration for UTI prevention. A sensitivity analysis was performed adjusting for sexual activity and current smoking status (which differed between per-protocol analysis participants). This also showed no difference in the primary outcome by application method (risk difference: 1.93 percentage points; 95% CI, −17.0 to 21.0 percentage points).

Table 1. Baseline Clinical and Demographic Characteristics by Randomization Arm

Characteristic	Intravaginal (n=57)	Periurethral (n=57)	*P*
Age (y)	69.9±8.4	72.2±9.2	.16
BMI (kg/m^2)	30.9±8.4	31.4±8.2	.77
Parity	2 [2–3]	2.5 [1.5–3]	.46
Baseline no. of UTIs			
Prior 6 mo	2 [2–3]	2 [2–3]	.69
Prior 12 mo	3 [2–4]	3 [3–4]	.69
Race			.72
Asian	1 (1.8)	1 (1.8)	
Black	5 (8.8)	2 (3.5)	
White	51 (89.5)	54 (94.7)	
Hispanic ethnicity	0 (0)	0 (0)	—
Current smoker	2 (3.5)	6 (10.5)	.27
Sexually active*	24 (42.9)	11 (19.6)	**<.01**
Medical history			
Hypertension	40 (70.2)	35 (61.4)	.43
Diabetes	14 (24.6)	13 (23.2)	.87
Fecal incontinence	7 (12.3)	6 (10.5)	.77
Neurogenic bladder	1 (1.8)	0 (0)	1.00
Chronic kidney disease	3 (5.3)	5 (8.8)	.49
History of kidney or bladder stones	10 (17.5)	9 (15.8)	.80
History of stroke	3 (5.3)	5 (8.8)	.72
History of DVT	1 (1.8)	0 (0)	1.00
Breast cancer	7 (12.3)	2 (3.5)	.16
Endometrial cancer	1 (1.8)	1 (1.8)	1.00
Medications at the time of enrollment			
Cranberry supplementation	4 (7.0)	8 (14.0)	.22
Probiotic for UTI prevention	3 (5.3)	4 (7.0)	1.00
Methenamine hippurate	4 (7.0)	5 (8.8)	1.00
D-mannose	7 (12.3)	6 (10.5)	.77
Anticholinergic OAB medication	8 (14.0)	8 (14.0)	1.00
β3-agonist	7 (12.3)	6 (10.5)	1.00
Immunosuppressant	5 (8.8)	1 (1.8)	.21
Frequent use of NSAIDs*,†	15 (26.8)	11 (19.3)	.38
Surgical history			
Hysterectomy	28 (49.1)	26 (45.6)	.85
Incontinence procedure	9 (15.8)	9 (15.8)	1.00
Prolapse repair	9 (16.1)	7 (12.3)	.79
Other characteristics			
Pessary use*	3 (5.4)	1 (1.8)	.62
Postvoid residual (mL)	21.5 [5–80]	28 [10–70]	.97

BMI, body mass index; UTI, urinary tract infection; DVT, deep vein thrombosis; OAB, overactive bladder; NSAID, nonsteroid anti-inflammatory drug.
Data are mean±SD, median [IQR], or n (%) unless otherwise specified.
Bold indicates statistical significance as defined as *P*<.05.
* Missing data: sexually active: intravaginal (n=56), periurethral (n=56); frequent use of NSAIDs: intravaginal (n=56), periurethral (n=57); pessary use: intravaginal (n=56), periurethral (n=55).
† Defined as at least every-other-day use.

Vaginal pH decreased in both arms at 6 months and did not differ by randomization arm (mean change: −1.2 intravaginal vs −0.8 periurethral, *P*=.30). The majority of participants reported feeling "much better" or "very much better" at 6 months (64.4% intravaginal vs 73% periurethral, *P*=.41) and agreed or strongly agreed with the statement, "I will continue to use the estrogen cream in the future." (78% intravaginal vs 75.6% periurethral, *P*=.65). Improvements in UDI-6 scores were similar between arms at 6 months (mean change: −5.03 points intravaginal vs −9.48 points periurethral, *P*=.18). Notably, there was minimal change in FSFI-6 scores with study drug use at 6 months (median change 0 intravaginal [IQR −3, 2] vs −1 periurethral [−3, 2], *P*=.50).

Participants reported that the estrogen cream was easy to use (86% intravaginal vs 85.4% periurethral, *P*=.93). There were few reported symptoms possibly

Table 2. Primary and Secondary Outcomes for the TAPER (Techniques of Applying Vaginal Estrogen for Prevention of Recurrent Urinary Tract Infections) Trial

Outcome	Intravaginal (n=57)	Periurethral (n=57)	*P*
Primary outcome			
% UTI free at 6 mo			
ITT analysis	30 (52.6)	29 (50.9)	.85
Per-protocol analysis*	26/50 (52.0)	22/45 (48.9)	.76
Secondary outcomes†			
Clinical and translational			
No. of UTIs at 6 mo	0 [0–1]	0 [0–2]	.53
Time to 1st UTI (d)	89 [55–147]	83 [45–121]	.08
Baseline vaginal pH	6 [5–7]	6 [5–7]	.55
Change in vaginal pH at 6 mo	−1.2±1.3	−0.8±1.5	.30
Patient-reported outcomes			
PGI-I ("much better" or "very much better") at 6 mo	29 (64.4)	27 (73.0)	.41
Baseline FSFI-6 score	15 [6–20]	6 [5–13]	**<.01**
Change in FSFI-6 score at 6 mo	0 [−3, 2]	−1 [−3, 2]	.50
Baseline UDI-6 score	40.3 [29.2–54.2]	43.5 [29.2–58.3]	.32
Change in UDI-6 score at 6 mo	−5.03±13.6	−9.48±16.9	.18
Patient experience with vaginal estrogen at 6 mo‡			
The estrogen cream is easy to use	43 (86.0)	35 (85.4)	.93
Applying the estrogen cream is messy	12 (24.0)	11 (26.8)	.78
Applying the estrogen cream is painful	2 (4.0)	0 (0)	.50
The estrogen cream irritates my tissues	3 (6.0)	0 (0)	.25
The estrogen cream is working to reduce my UTIs	38 (76.0)	31 (75.6)	.97
I am satisfied with the estrogen cream	39 (78.0)	32 (78.1)	1.00
I will continue to use the estrogen cream in the future	39 (78.0)	31 (75.6)	.65
Study drug adverse effects (within 1st 3 mo of use)			
Vulvar swelling	3 (6.0)	0 (0)	.25
Vulvar itching	11 (22.0)	3 (7.3)	.05
Vaginal itching	12 (24.0)	1 (2.4)	**<.01**
Vaginal discharge	4 (8.0)	1 (2.4)	.37
Vaginal burning	5 (10.0)	1 (2.4)	.22
Vaginal pain	3 (6.0)	1 (2.4)	.62
Vaginal bleeding	2 (4.0)	1 (2.4)	1.00
Breast tenderness	4 (8.0)	2 (4.9)	.69
Breast swelling	1 (2.0)	1 (2.4)	1.00

UTI, urinary tract infection; ITT, intention-to-treat; PGI-I, Patient Global Impression of Improvement; UDI-6, Urogenital Distress Inventory, Short Form; FSFI-6, Female Sexual Function Index-6.

Data are n (%), n/N (%), median [IQR], or mean±SD unless otherwise specified.

Bold indicates significance (*P*<.05).

* At least 50% compliance with randomized study drug application method.

† Available data at 6 months: vaginal pH: n=34 intravaginal, n=27 periurethral; PGI-I: n=45 intravaginal; n=37 periurethral; FSFI-6: n=43 intravaginal; n=38 periurethral; UDI-6: n=48 intravaginal; n=40 periurethral; estrogen experience questionnaire: n=50 intravaginal; n=41 periurethral; study drug adverse effects (at 3 months): n=50 intravaginal; n=41 periurethral.

‡ Comparison of positive responses (agree, strongly agree) on 5-point Likert scale for the following statements. Data analyzed using χ^2 analysis and presented as n (%).

associated with use of the study drug, although participants who applied estrogen intravaginally were more likely to experience vaginal itching after 3 months of use (24% vs 2.4%, *P*<.01). This association was no longer significant after 6 months of study drug use (14% intravaginal vs 9.8% periurethral, *P*=.54). The only participants who agreed with the statements, "Applying the estrogen cream is painful" (n=2) and "The estrogen cream irritates my tissues" (n=3) were in the intravaginal arm, although the difference by application method did not reach statistical significance.

DISCUSSION

Among postmenopausal individuals with laboratory-documented recurrent UTI, periurethral administration of 0.5 g estradiol cream was noninferior to intravaginal administration of 1.0 g estradiol cream for the prevention of UTI over 6 months. Both strategies were viewed favorably by participants, despite an increased prevalence of vaginal itching at 3 months of use associated with intravaginal estradiol application.

Despite numerous randomized trials confirming that intravaginal estrogen is effective for UTI

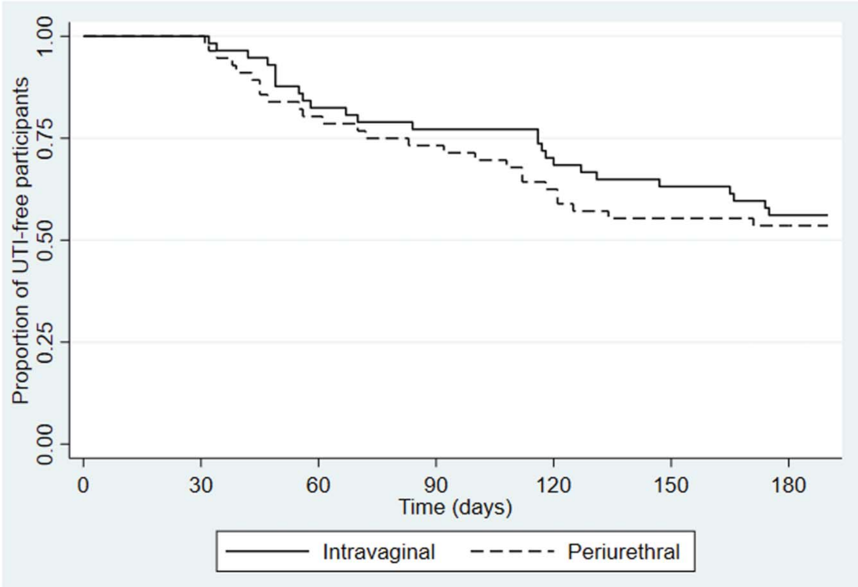

Fig. 2. Kaplan-Meier curve for urinary tract infection (UTI)–free survival over 6 months. No difference in UTI-free survival between groups on adjusted Cox proportional hazards model that controls for baseline sexual activity (HR 0.93; 95% CI, 0.53–1.63).

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prevention,[18–20] there are a scarcity of published data comparing alternative routes of vaginal estrogen application for the prevention of UTI. Application of a lower volume of estrogen cream to the vulva or vagina has been described by the American Urogynecologic Society in patient pamphlets but is not included in U.S. Food and Drug Administration brochures for vaginal estrogen creams.[21] In one of the few published studies on periurethral application of estrogen cream, Maiti et al[22] compared periurethral application of estrogen with tamsulosin 0.4 mg for the treatment of lower urinary tract symptoms in perimenopausal individuals over a 6-week period and found periurethral estrogen to be less effective. A pilot study of 17 biological females who used 0.5 g vaginal estradiol cream studied digital application to the inside of the vagina for 4 months and reported improved vaginal maturation index and symptoms of vulvovaginal atrophy but did not examine effects on UTIs.[23]

Periurethral tissues are known to be estrogen-sensitive.[24] Vaginal estrogen increases periurethral blood flow and vascularization, especially if applied to the outer one-third of the vagina.[25,26] Our study shows that decreases in vaginal pH in the distal, posterior vagina occur with periurethral administration of estradiol, in a similar fashion as intravaginal administration. Notably, periurethral application of estrogen cream requires a lower amount of medication administered to the outer vagina, resulting in less absorption.[26] This may be viewed favorably by individuals concerned about exogenous estrogen exposure and by those concerned about medication costs, particularly because topical estrogen is often prescribed for long-term use.

Previous studies have found that the standard administration of vaginal estrogen cream with an intravaginal applicator takes on average 5.08 minutes to apply.[27] Two-thirds of individuals find vaginal estrogen creams to be “messy,” and 48% report that washing the intravaginal applicator is inconvenient.[10,28] Our study found no difference in patient satisfaction, ease of use, or messiness with estradiol cream by administration method. This may be due to the fact that participants in the periurethral arm were instructed to use an applicator to ensure standard “pea-sized” doses. Real-world usage of this method would not require patients to use an applicator. Patient experiences did differ in that vaginal itching in the first 3 months of use occurred more frequently in the intravaginal arm. Vaginal itching is a recognized adverse effect of topical estrogen therapy, likely due to irritation of atrophic tissues with initial application.[29]

Limitations include that the trial was not powered to detect smaller differences in application method effectiveness under a narrower noninferiority margin. Nevertheless, the TAPER trial provides evidence of clinical equipoise between periurethral and intravaginal application of estradiol for UTI prevention, and therefore offers a strong basis for a larger, confirmatory trial with a placebo arm. Patient experience with the study drug was assessed using a nonvalidated

Authors' Data Sharing Statement

- Will individual participant data be available (including data dictionaries)? *Individual deidentified participant data, including the data dictionary, are available by request to the corresponding author. Data access will require a data use agreement.*
- What data in particular will be shared? *Clinical data from this trial may be shared with a data use agreement.*
- What other documents will be available? *No other documents are available.*
- When will data be available (start and end dates)? *Data can be requested at any time after publication of this article.*
- By what access criteria will data be shared (including with whom, for what types of analyses, and by what mechanism)? *Data may be shared with any entity on formal request after review by the corresponding author and execution of a data use agreement with the sponsoring institution (University of Pittsburgh).*

survey, which would benefit from further research. In addition, other nonantibiotic prophylactic therapy, such as methenamine hippurate, D-mannose, probiotics, and cranberry supplementation, were allowed to be initiated during study enrollment, which could influence study results, but these data were not available for inclusion in sensitivity analyses. Finally, adherence with the study drug was based on self-report. Although study drug weights were collected when possible, many participants did not return their tubes of study product at follow-up, which precluded a reliable objective measure of study product adherence.

Strengths of the study include the strict definition of recurrent UTIs and postmenopausal status for inclusion criteria. Participant characteristics were comparable after randomization, except for sexual activity, which was addressed through a sensitivity analysis. In addition, a unique aspect of the study was the option for remote enrollment and participation, which enabled recruitment of individuals with transportation challenges and from rural communities, thereby improving the generalizability of study findings.

In conclusion, periurethral application of a lower dose of estrogen cream was found to be noninferior to intravaginal application in UTI prevention among postmenopausal individuals with recurrent UTIs. Health care professionals may consider recommending this alternative application method to patients who find the use of an intravaginal applicator to be unacceptable or technically challenging (eg, due to arthritis, obesity, or vaginal stenosis).

REFERENCES

1. Jung C, Brubaker L. The etiology and management of recurrent urinary tract infections in postmenopausal women. Climacteric 2019;22:242–9. doi: 10.1080/13697137.2018.1551871
2. Peck J, Shepherd JP. Recurrent urinary tract infections: diagnosis, treatment, and prevention. Obstet Gynecol Clin North Am 2021;48:501–13. doi: 10.1016/j.ogc.2021.05.005
3. Eriksson I, Olofsson B, Gustafson Y, Fagerström L. Older women's experiences of suffering from urinary tract infections. J Clin Nurs 2014;23:1385–94. doi: 10.1111/jocn.12422
4. Dieter AA, Mueller MG, Andy UU, Kirby AC, Dwarica DS, Amundsen CL, et al. Baseline characteristics, evaluation, and management of women with complaints of recurrent urinary tract infections. Female Pelvic Med Reconstr Surg 2021;27:275–80. doi: 10.1097/SPV.0000000000001065
5. Sihra N, Goodman A, Zakri R, Sahai A, Malde S. Nonantibiotic prevention and management of recurrent urinary tract infection. Nat Rev Urol 2018;15:750–76. doi: 10.1038/s41585-018-0106-x
6. O'Hanlon DE, Moench TR, Cone RA. Vaginal pH and microbicidal lactic acid when lactobacilli dominate the microbiota. PLoS One 2013;8:e80074. doi: 10.1371/journal.pone.0080074
7. Jung CE, Estaki M, Chopyk J, Taylor BC, Gonzalez A, McDonald D, et al. Impact of vaginal estrogen on the urobiome in postmenopausal women with recurrent urinary tract infection. Female Pelvic Med Reconstr Surg 2022;28:20–6. doi: 10.1097/SPV.0000000000001051
8. Shulman LP, Portman DJ, Lee WC, Balu S, Joshi AV, Cobden D, et al. A retrospective managed care claims data analysis of medication adherence to vaginal estrogen therapy: implications for clinical practice. J Womens Health (Larchmt) 2008;17:569–78. doi: 10.1089/jwh.2007.0407
9. Mattsson LÅ, Ericsson Å, Bøgelund M, Maamari R. Women's preferences toward attributes of local estrogen therapy for the treatment of vaginal atrophy. Maturitas 2013;74:259–63. doi: 10.1016/j.maturitas.2012.12.004
10. Minkin MJ, Maamari R, Reiter S. Improved compliance and patient satisfaction with estradiol vaginal tablets in postmenopausal women previously treated with another local estrogen therapy. Int J Womens Health 2013;5:133–9. doi: 10.2147/IJWH.S41897
11. Buck ES, Lukas VA, Rubin RS. Effective prevention of recurrent utis with vaginal estrogen: pearls for a urological approach to genitourinary syndrome of menopause. Urology 2021;151:31–6. doi: 10.1016/j.urology.2020.05.058
12. Zuo SW, Bradley M, Mowers EE, Zyczynski HM, Hillier SL, Ackenbom MF. Periurethral versus intravaginal application of vaginal estrogen for the prevention of urinary tract infections in postmenopausal women: design of a randomized non-inferiority trial. Contemp Clin Trials 2025;153:107893. doi: 10.1016/j.cct.2025.107893
13. Bodenheimer T. Teach-Back: a simple technique to enhance patients' understanding. Fam Pract Manag 2018;25:20–2.
14. Ayton RA, Darling GM, Murkies AL, Farrell EA, Weisberg E, Selinus I, et al. A comparative study of safety and efficacy of continuous low dose oestradiol released from a vaginal ring compared with conjugated equine oestrogen vaginal cream in the treatment of postmenopausal urogenital atrophy. Br J Obstet Gynaecol 1996;103:351–8. doi: 10.1111/j.1471-0528.1996.tb09741.x

15. Perrotta C, Aznar M, Mejia R, Albert X, Ng CW. Oestrogens for preventing recurrent urinary tract infection in postmenopausal women. The Cochrane Database of Systematic Reviews 2008, Issue 2. Art. No.: CD005131. doi: 10.1002/14651858.CD005131.pub2
16. Svatek R, Roche V, Thornberg J, Zimmern P. Normative values for the American Urological Association Symptom Index (AUA-7) and short form Urogenital Distress Inventory (UDI-6) in patients 65 and older presenting for non-urological care. Neurourol Urodyn 2005;24:606–10. doi: 10.1002/nau.20189
17. Isidori AM, Pozza C, Esposito K, Giugliano D, Morano S, Vignozzi L, et al. Development and validation of a 6-item version of the Female Sexual Function Index (FSFI) as a diagnostic tool for female sexual dysfunction. J Sex Med 2010;7:1139–46. doi: 10.1111/j.1743-6109.2009.01635.x
18. Eriksen B. A randomized, open, parallel-group study on the preventive effect of an estradiol-releasing vaginal ring (Estring) on recurrent urinary tract infections in postmenopausal women. Am J Obstet Gynecol 1999;180:1072–9. doi: 10.1016/s0002-9378(99)70597-1
19. Raz R, Stamm WE. A controlled trial of intravaginal estriol in postmenopausal women with recurrent urinary tract infections. N Engl J Med 1993;329:753–6. doi: 10.1056/NEJM199309093291102
20. Ferrante KL, Wasenda EJ, Jung CE, Adams-Piper ER, Lukacz ES. Vaginal estrogen for the prevention of recurrent urinary tract infection in postmenopausal women: a randomized clinical trial. Female Pelvic Med Reconstr Surg 2021;27:112–7. doi: 10.1097/SPV.0000000000000749
21. American Urogynecologic Society. Vaginal estrogen therapy. Accessed June 15, 2026. https://www.voicesforpfd.org/wp-content/uploads/2025/02/Vaginal_Estrogen_Therapy.pdf
22. Maiti K, Jaiswal A, Pal DK. A comparative study of alpha-1a blockers (tamsulosin) versus estrogens in the treatment of lower urinary tract symptoms in perimenopausal females. Indian J Pharmacol 2020;52:6–9. doi: 10.4103/ijp.IJP_545_18
23. Illston JD, Wheeler TL, Parker CR, Conner MG, Burgio KL, Goode PS, et al. Low-dose 17-β-estradiol cream for vaginal atrophy in a cohort without prolapse: serum levels and vaginal response including tissue biomarkers associated with tissue remodeling. Maturitas 2015;81:475–9. doi: 10.1016/j.maturitas.2015.05.010
24. Skala CE, Petry IB, Albrich SB, Puhl A, Naumann G, Koelbl H. The effect of hormonal status on the expression of estrogen and progesterone receptor in vaginal wall and periurethral tissue in urogynecological patients. Eur J Obstet Gynecol Reprod Biol 2010;153:99–103. doi: 10.1016/j.ejogrb.2010.07.006
25. Kobata SA, Girão MJBC, Baracat EC, Kajikawa M, Di Bella V Jr, Sartori MGF, et al. Estrogen therapy influence on periurethral vessels in postmenopausal incontinent women using Dopplervelocimetry analysis. Maturitas 2008;61:243–7. doi: 10.1016/j.maturitas.2008.07.015
26. Cicinelli E, Di Naro E, De Ziegler D, Matteo M, Morgese S, Galantino P, et al. Placement of the vaginal 17beta-estradiol tablets in the inner or outer one third of the vagina affects the preferential delivery of 17beta-estradiol toward the uterus or periurethral areas, thereby modifying efficacy and endometrial safety. Am J Obstet Gynecol 2003;189:55–8. doi: 10.1067/mob.2003.341
27. Minkin MJ, Maamari R, Reiter S. Postmenopausal vaginal atrophy: evaluation of treatment with local estrogen therapy. Int J Womens Health 2014;6:281–8. doi: 10.2147/IJWH.S57900
28. The NAMS 2020 GSM Position Statement Editorial Panel. The 2020 genitourinary syndrome of menopause position statement of the North American Menopause Society. Menopause 2020;27:976–92. doi: 10.1097/GME.0000000000001609
29. U.S. Food and Drug Administration. FDA-approved label for ESTRACE Cream. Accessed March 28, 2026. https://www.accessdata.fda.gov/drugsatfda_docs/label/2022/086069Orig1s026lbl.pdf

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